

Weighted Reflection Channels and a Natural-Boundary Circle

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Abstract

The orbit-resolved reflection Euler product of the full Hénon horseshoe admits a two-variable scalar-channel continuation. We classify the complete complex singular geometry on every positive weight fiber. The m th channel has coefficient

$$c_m = \frac{1}{m} \prod_{\substack{p|m \\ p \text{ odd}}} (1 - p) \neq 0$$

and poles

$$\alpha_{m,k}(q) = (1 + q^{2m})^{-1/(2m)} e^{\pi i k/m}, \quad 0 \leq k < 2m.$$

For every $q > 0$, the channel radii increase strictly to $L(q) = \min(1, q^{-1})$. Each listed point is an exponential essential singularity of the Euler product, while the angular mesh becomes dense. It follows that $|z| = L(q)$ is a natural boundary for the exact unrenormalized punctured continuation. For $q > 1$ the limiting circle lies strictly inside the unit disk. The conclusion is deliberately object-specific: an all-channel counterterm changes the function, and no weighted Lind source, transfer operator, arithmetic trace, or Route-B claim is obtained.

1 From the first boundary to the full complex divisor

The area-preserving Hénon map on the frozen horseshoe is conjugate to the full two-shift in the regime used throughout this programme [2]. For the cross-axis observable $\chi = \mathbf{1}\{s_{-1} = s_1\}$, the earlier orbit-resolved calculation gives the odd reflection polynomial

$$F_{2r+1}(q) = 2q(1 + q^2)^r$$

and the primitive dilation–Möbius inversion

$$E_n(q) = \sum_{d|n} \mu(d) F_{n/d}(q^d).$$

The associated product is

$$\mathcal{Z}_{\text{orb}}(z, q) = \prod_{\substack{n \geq 1 \\ n \text{ odd}}} \prod_{\omega \in A_n} (1 - z^n q^{S_n \chi(\omega)})^{-1}. \quad (1)$$

Its first positive convergence boundary is $(1 + q^2)^{-1/2}$, where it is essentially singular.

The point of the present paper is that the first boundary is only the first member of a rigid complex divisor. The weighted channel regrouping reads

$$\log \mathcal{Z}_{\text{orb}}(z, q) = \sum_{m \geq 1} c_m \Psi_m(z, q), \quad \Psi_m(z, q) = \frac{2(qz)^m}{1 - (1 + q^{2m})z^{2m}}, \quad (2)$$

initially in the disk of the Euler product and then on the complement of its locally finite polar set. Here

$$c_m = \frac{1}{m} \sum_{\substack{d|m \\ d \text{ odd}}} d \mu(d) = \frac{1}{m} \prod_{\substack{p|m \\ p \text{ odd}}} (1 - p). \quad (3)$$

In particular no channel disappears.

2 Strict separation of channel radii

For $m \geq 1$ and $q > 0$, put

$$\rho_m(q) = (1 + q^{2m})^{-1/(2m)}. \quad (4)$$

Theorem 2.1 (Moving-radius ladder). *For every fixed $q > 0$,*

$$\rho_1(q) < \rho_2(q) < \rho_3(q) < \dots$$

and

$$\lim_{m \rightarrow \infty} \rho_m(q) = L(q) := \min(1, q^{-1}). \quad (5)$$

Moreover $\rho_m(q) = q^{-1} \rho_m(q^{-1})$.

Proof. For $p > 0$, let

$$N_p(q) = \|(1, q)\|_p = (1 + q^p)^{1/p}.$$

The finite-dimensional ℓ^p norm of a vector with two nonzero coordinates is strictly decreasing in p . Since $\rho_m(q) = N_{2m}(q)^{-1}$, strict increase follows. The standard $p \rightarrow \infty$ limit is $N_p(q) \rightarrow \max(1, q)$, which proves (5). Factoring q^{2m} from (4) gives the final identity. $\square$

Strictness is important: singularities belonging to different channel indices never collide on a positive-weight fiber. This rules out cross-channel cancellation before any local expansion is computed.

3 Every complex channel root is essential

The denominator in (2) has exactly the roots

$$\alpha_{m,k}(q) = \rho_m(q) e^{\pi i k/m}, \quad 0 \leq k < 2m. \quad (6)$$

Proposition 3.1 (Exact local coefficient). *Fix $q > 0$, $m \geq 1$, and $0 \leq k < 2m$. With $v = 1 - z/\alpha_{m,k}(q)$,*

$$\log \mathcal{Z}_{\text{orb}}(z, q) = \frac{c_m (-1)^k q^m}{m \sqrt{1 + q^{2m}} v} + G_{m,k,q}(z), \quad (7)$$

where $G_{m,k,q}$ is holomorphic near $\alpha_{m,k}(q)$.

Proof. The root identity gives

$$\alpha_{m,k}(q)^m = \frac{(-1)^k}{\sqrt{1+q^{2m}}}.$$

Writing $z = \alpha_{m,k}(q)(1-v)$, we obtain

$$1 - (1+q^{2m})z^{2m} = 2mv + O(v^2)$$

and

$$2(qz)^m = \frac{2(-1)^k q^m}{\sqrt{1+q^{2m}}} + O(v).$$

Their quotient is the principal term in (7). By theorem 2.1, every other channel has its poles on a different circle and is therefore holomorphic locally. Normal convergence of the remaining tail supplies $G_{m,k,q}$. $\square$

Corollary 3.2 (Full essential divisor). *Every point in (6) is an exponential essential singularity of the continuation of $\mathcal{Z}_{\text{orb}}(\cdot, q)$.*

Proof. The coefficient in (7) is nonzero by (3). Exponentiating a nonzero simple pole gives an essential singularity. $\square$

4 The limiting circle is a natural boundary

For fixed m , the arguments in (6) form the uniform grid

$$0, \frac{\pi}{m}, \dots, \frac{(2m-1)\pi}{m}.$$

Its maximum gap is π/m .

Theorem 4.1 (Natural-boundary circle). *For every $q > 0$, no point of the circle*

$$|z| = L(q) = \min(1, q^{-1})$$

has a neighborhood to which the explicit unrenormalized continuation $\mathcal{Z}_{\text{orb}}(\cdot, q)$ extends meromorphically. Thus this circle is a natural boundary for that punctured continuation.

Proof. Let $z_0 = L(q)e^{i\theta}$. Choose a grid angle $\pi k_m/m$ at distance at most $\pi/(2m)$ from θ . By theorem 2.1,

$$\alpha_{m,k_m}(q) = \rho_m(q)e^{\pi i k_m/m} \longrightarrow z_0.$$

Every term of this sequence is an essential singularity by corollary 3.2, and the radii are distinct. If a meromorphic continuation existed on a neighborhood of z_0 , that neighborhood would contain infinitely many of these nonmeromorphic isolated singularities accumulating at its interior point z_0 . This contradicts the local structure of a meromorphic function [1]. $\square$

The theorem has three regimes. When $0 < q < 1$, the natural boundary is the unit circle. At $q = 1$, it is the unweighted boundary already approached by the positive P72 ladder. When $q > 1$, it moves inward to $|z| = q^{-1}$. The last behavior is not a numerical phase transition; it is the exact ℓ^∞ limit in theorem 2.1.

5 Renormalization and operator firewalls

The natural-boundary terminology in theorem 4.1 refers to analytic continuation of the fixed function (1) through the exact channel continuation (2). It does not say that the prescribed divisor cannot be subtracted. An all-channel exponential counterterm can cancel it precisely because the counterterm contains the same complete ledger. The resulting function is a different object.

Likewise, the theorem does not identify a weighted analogue of the full reverse-shift Lind zeta of Kim et al. [3]. No source formula currently supplies such a comparison for $q \neq 1$. Nor is the channel continuation itself a transfer determinant: a parameter-dependent diagonal operator can be engineered after the fact, while the direct sum of source-native orbit blocks has a separate compactness gate. Those two notions are compared in the next paper.

In particular, the channel index m has no intrinsic rational-prime meaning, and no prime-power amplitude, explicit formula, self-adjoint operator, zeta-zero correspondence, or Route-B result follows.

6 Executable certificate

The main certificate reconstructs forty channels on each of the fibers $q = 1/2, 1, 2$, using high-precision decimal radii to preserve strictness near the limiting circle. It checks every complex root numerically, records the principal coefficient and angular gap, locks nine P70/P72/P75 dependencies, and rejects 24 claim mutations. A separate program reconstructs 192 rows on the fibers $q = 0.4, 1, 2.5$ without importing the main module. Eleven unit tests pass in normal and optimized modes. These finite checks certify the implementation; theorems 2.1 and 4.1 prove the all-channel claims.

References

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